TASK-3

The relay should turned on when the ultrasonicsensor measures a distance within 30 cm (Threshold) .

The Wiring Guide

Ultrasonic Sensor (HC-SR04)

	ESP32 WROOM Pin	Notes
VCC	5V / VIN	Power for the sensor
GND	GND	Ground
Trig	GPIO 5	Output from ESP32 to trigger pulse
Echo	GPIO 18	Input to ESP32 (Use voltage divider here!)

Relay Module	ESP32 WROOM Pin	Notes
VCC	5V / VIN or 3.3V	Depending on your relay board spec
GND	GND	Ground
IN	GPIO 19	Controls the relay switch

MicroPython Code

```
import machine
import time

# --- Configuration ---
TRIGGER_PIN = 5
ECHO_PIN = 18
RELAY_PIN = 19
THRESHOLD_CM = 30.0 # Distance threshold in cm

# --- Pin Initialization ---
trig = machine.Pin(TRIGGER_PIN, machine.Pin.OUT)
echo = machine.Pin(ECHO_PIN, machine.Pin.IN)
relay = machine.Pin(RELAY_PIN, machine.Pin.OUT)

# Ensure relay starts OFF
# Note: Active-Low relays turn ON when output is 0. Active-High turn ON when 1.
# If your relay behavior is inverted, change this to 0.
relay.value(0)

def get_distance():
```

```
    """MeasuresdistanceusingtheHC-SR04ultrasonic sensor."""
    # Ensure trigger is low
    trig.value(0)
    time.sleep_us(2)

    # Send a 10 microsecond pulse to trigger
    trig.value(1)
    time.sleep_us(10)
    trig.value(0)

    # Measure the duration of the echo pulse
    duration=machine.time_pulse_us(echo,1,30000) # 30ms timeout

    if duration < 0:

        # If timeout occurs (return value -1 or -2)
        return None

    # Calculate distance in cm
    # Speed of sound is ~343 m/s or 0.0343 cm/us.
    # Pulse goes there and back, so we divide by 2.
    distance = (duration * 0.0343) / 2
    return distance

print("Ultrasonic Relay Controller Initialized...")

# --- Main Loop ---
while True:
    dist = get_distance()

    if dist is_not None:
        print(f"Distance: {dist:.2f} cm")

        # Check if an object is within the threshold
        if dist <= THRESHOLD_CM:
            print("Target detected! Turning Relay ON.")
            relay.value(1)#Setto0ifusinganActive-Low relay module
        else:
            print("Clear. Turning Relay OFF.")
            relay.value(0)#Setto1ifusinganActive-Low relay module
    else:
        print("Out of range or sensor error.")

    #Wait500msbeforethenextreadingtoavoidflooding the sensor
    time.sleep(0.5)
```

NOTE:

THRESHOLD_CM =30 (Add this in the code)

RESULT: The ultrasonic sensor turns off when the object is detected beyond 30cm